Discrete vs. Continuous: Modeling the Beer Game in StochSD and Excel

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“Understanding systems before we change them”

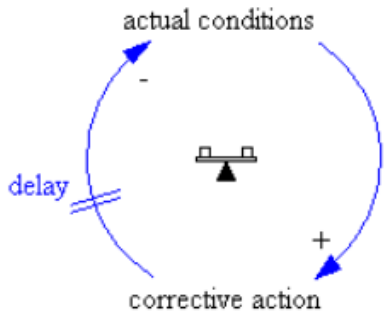
Abstract

This paper compares two models of the Beer Game — one implemented in Excel and the other in StochSD — to examine the impact of order discreteness on the bullwhip effect. It is shown that introducing minimum order quantities amplifies order fluctuations. The StochSD model does not accumulate orders, which smooths the dynamics, whereas Excel allows explicit control over the computation sequence and accounts for unfilled orders, yielding more realistic outcomes. The conclusion suggests that spreadsheets are preferable to classical system dynamics environments for modeling supply chains with discrete lot sizes.

1. Introduction

The "balancing process with delay" archetype describes systems in which the response to an action occurs with a time lag. If agents do not take this delayed feedback into account, they may underestimate or overestimate the actions needed to achieve their goals. A diagram of the archetype is shown in Figure 1.

Figure 1. Diagram of the "balancing process with delay" archetype



A common everyday example of this archetype is adjusting the water temperature in a shower: the water temperature changes with a delay relative to the movement of the tap.

In a business context, the classic example of the "balancing process with delay" archetype is the Beer Game. The game represents a supply chain with misaligned processes. Due to a lack of information, all groups have an incomplete picture of real demand. Each

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group can control only its own part of the supply chain, yet its ordering decisions — too many or too few — can significantly affect the entire chain. Thus, the actions of one group largely depend on the decisions of others, and vice-versa.

2. Description of the Beer Game

The common scheme of the Beer Game (Figure 2) represents a supply chain consisting of four groups:

1) Retail group – receives orders from end customers;
2) Wholesale group – supplies goods to the Retail group;
3) Regional Warehouse group – supplies goods to the Wholesale group;
4) Factory group – produces the goods (beer) and supplies them to the Regional Warehouse group.

For simplicity, the first step — waiting for raw materials at the factory — is not considered.

Figure 2: Классическая схема пивной игры

The main problem of the supply chain is the delays in time to fill orders and delivery time between each group. Each group tries to maintain a desired warehouse supply level, but due to delays and information distortion, the bullwhip effect arises: small fluctuations in demand at the Retail level turn into large fluctuations in orders at the Regional Warehouse level.

Let’s examine the Retail group in more detail. The Retail group has a warehouse where beer barrels are stored, end customers, and a supplier (Wholesale group) who delivers new batches to the Retail group warehouse, but not immediately — only several weeks after an order is placed.

The goal of the Retail group is to keep beer in stock while avoiding excessive inventory. Order decisions are made based on local information:

1) Amount of outgoing goods during the current week;
2) Current warehouse balance;
3) Quantity of the beer already ordered but not yet delivered;
4) Desired warehouse supply level – a safety buffer against delivery delays.

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If the current supply falls below the desired level, the Retail group increases the order to the Wholesale group, even if a recent order has already been placed. The same logic is applied sequentially by the Wholesale group, the Regional Warehouse group, and finally the Factory group. The Factory group is responsible only for production; the produced goods move in the opposite direction: Factory group → Regional Warehouse group → Wholesale group → Retail group.

For the purpose of modeling, it is assumed that the warehouses of the Retail group, Wholesale group, and Regional Warehouse group have infinite capacity.

3. Beer Game Model Implementation

3.1. Excel Model

Based on the common Beer Game scheme (Figure 2), a numerical model was developed in Excel. The Retail group and the Factory group, as the first and last groups in the chain, have their own specific features, whereas the Wholesale group and the Regional Warehouse group operate identically. Each group is examined separately.

Time unit: Week

Main parameters for the Retail group:
- Initial supply Retail;
- Delivery time to Retail;
- Average amount of sales Retail;
- Time to fill order for Retail;
- Desired supply Retail.

Main parameters for the Wholesale group:
- Initial supply Wholesale;
- Delivery time to Wholesale;
- Time to fill order for Wholesale;
- Desired supply Wholesale.

Main parameters for the Regional Warehouse group:
- Initial supply Regional Warehouse;
- Delivery time to Regional Warehouse;
- Time to fill order for Regional Warehouse;
- Desired supply Regional Warehouse.

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Main parameters for the Factory group:

- Initial supply Factory;
- Average production Factory;
- Max supply Factory.

Logic of the groups:

- Retail group: The Wholesale group delivers the order; the incoming goods are added to the warehouse. Sales are made from the warehouse supply (every 4 weeks, demand increases by 200 units). After sales, the warehouse empties, and the Retail group places a new order with the Wholesale group, based on current supply and desired supply (outgoing orders of the Retail group become incoming orders for the Wholesale group). Retail group orders are placed in a queue; after the Time to fill order for Retail, the Wholesale group takes the order for fulfillment. The Wholesale group then ships the order to the Retail group, and the goods arrive at the Retail group's warehouse after the Delivery time to Retail.

- Wholesale group: The Regional Warehouse group delivers the Wholesale group's order; the incoming goods are added to the warehouse. Sales to the Retail group are made from the warehouse (if the warehouse is empty, there are no sales). After selling to the Retail group, the Wholesale group's warehouse empties, and the Wholesale group places an order with the Regional Warehouse group based on its own supply and desired supply. Orders are queued; after the Time to fill order for Wholesale, the Regional Warehouse group takes the order for fulfillment. The Regional Warehouse group ships the order to the Wholesale group, and the goods arrive after the Delivery time to Wholesale.

- Regional Warehouse group: The Factory group delivers the Regional Warehouse group's order; the incoming goods are added to the Regional Warehouse group's warehouse. Sales to the Wholesale group are made from the warehouse. After selling to the Wholesale group, the Regional Warehouse group's warehouse empties, and the Regional Warehouse group places an order with the Factory group based on its own supply and desired supply. Orders are queued; after the Time to fill order for Regional Warehouse, the Factory group takes the order for fulfillment. The Factory group ships the order to the Regional Warehouse group, and the goods arrive after the Delivery time to Regional Warehouse.

- Factory group: The Factory group produces between 400 and 1200 units per week, depending on demand and warehouse occupancy. It takes orders from the Regional Warehouse group's order queue; during the Time to fill order for Regional Warehouse, the Factory group prepares the order for shipment. The Factory group ships the order to the

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Regional Warehouse group, and the goods arrive after Delivery time to Regional Warehouse.

Figure 3 shows the movement of goods and orders for the first 30 weeks for all four groups.

Example of order flow:

Step 1. Retail group's order – 700 units. After the Time to fill order for Retail (1 week), this order enters the Wholesale group's order queue. After the Delivery time to Retail (2 weeks), the goods arrive at the Retail group's warehouse.

Step 2. Retail group's order – 1700 units, consisting of 700 units from a previous unfulfilled order (already shipped by the Wholesale group but not yet received by the Retail group) plus 1000 new units. This order enters the Wholesale group's order queue. The Wholesale group has only 500 units in supply, which are immediately shipped to the Retail group. The remaining part of the order will be shipped later, after the Wholesale group replenishes its supply to the desired level of 1500 units.

The Wholesale group's order to the Regional Warehouse group now accumulates to 2800 units: 1300 units from previous orders + 1500 new units. After the Time to fill order for Wholesale (2 weeks), this order enters the Regional Warehouse group's order queue. However, the Regional Warehouse group has only 200 units in supply, which are shipped to the Wholesale group. Thus, 2600 units remain unshipped. To fulfill them, the Regional Warehouse group must first replenish its supply to the desired level of 2100 units.

The Regional Warehouse group's order to the Factory group amounts to 5200 units: 2100 new units + 3100 units from previous orders. After the Time to fill order for Regional Warehouse (3 weeks), the order enters the Factory group queue, where 400 units of previously unshipped goods are added, resulting in 5600 units queued at the Factory group. The Factory group operates at full capacity (1200 units per week), but orders continue to accumulate.

Result: The Factory group ships 1200 units per week to the distributor. These 1200 units are immediately forwarded: the Regional Warehouse group ships them to the Wholesale group, and the Wholesale group ships them to the Retail group.

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Figure 3. Beer Game calculation in Excel

Model Excel_1 w/o limits				Incoming goods Retail, Unit/ Week	Warehouse Retail, Unit	Outgoing goods Retail, Unit/ Week	Orders Retail, Unit/ Week	Incoming goods Wholesale, Unit/ Week	Warehouse Wholesale, Unit	Placed orders Retail, Unit	Outgoing goods Wholesale, Unit/ Week	Orders Wholesale, Unit/ Week	Incoming goods RWH, Unit/ Week	Warehouse RWH, Unit	Outgoing orders Wholesale, Unit/ Week	Outgoing goods RWH, Unit/ Week	Orders RWH, Unit/ Week	Production Factory, Unit/ Week	Warehouse Factory, Unit	Outgoing orders RWH, Unit/ Week	Outgoing goods Factory, Unit/ Week	
TIME STEP	1	Week	0	0,00	900,00	0,00	0,00	0,00	1200,00	0,00	0,00	0,00	0,00	1800,00	0,00	0,00	0,00	0,00	2400,00	0,00	0,00	0
			1	0,00	900,00	300,00	700,00	0,00	1200,00	0,00	0,00	300,00	0,00	1800,00	0,00	0,00	300,00	400,00	2400,00	0,00	0,00	1
Initial supply Retail	900	Unit	2	0,00	600,00	300,00	1700,00	0,00	1200,00	700,00	700,00	1300,00	0,00	1800,00	0,00	0,00	600,00	400,00	2800,00	0,00	0,00	2
Delivery time to Retail	2	Week	3	0,00	300,00	300,00	3000,00	0,00	500,00	1700,00	500,00	2800,00	0,00	1800,00	300,00	300,00	1200,00	400,00	3200,00	0,00	0,00	3
Average amount of sales Retail	300	Unit/ Week	4	700,00	700,00	500,00	4100,00	0,00	0,00	4200,00	0,00	4300,00	0,00	1500,00	1300,00	1300,00	3100,00	400,00	3600,00	300,00	300,00	4
Time to fill oder for Retail	1	Week	5	500,00	700,00	300,00	5000,00	0,00	0,00	8300,00	0,00	5800,00	0,00	200,00	2800,00	200,00	5200,00	400,00	3700,00	600,00	600,00	5
Desired supply Retail	1300	Unit	6	0,00	400,00	300,00	6200,00	300,00	300,00	13300,00	300,00	7300,00	0,00	0,00	6900,00	0,00	7300,00	400,00	3500,00	1200,00	1200,00	6
			7	0,00	100,00	100,00	7500,00	1300,00	1300,00	19200,00	1300,00	8800,00	0,00	0,00	12700,00	0,00	9400,00	400,00	2700,00	3100,00	2700,00	7
Initial supply Wholesale	1200	Unit	8	300,00	300,00	300,00	8800,00	200,00	200,00	25400,00	200,00	10300,00	300,00	300,00	20000,00	300,00	11500,00	800,00	400,00	5600,00	400,00	8
Delivery time to Wholesale	3	Week	9	1300,00	1300,00	300,00	9100,00	0,00	0,00	34000,00	0,00	11800,00	600,00	600,00	28500,00	600,00	13600,00	1200,00	800,00	12500,00	800,00	9
Time to fill oder for Wholesale	2	Week	10	200,00	1200,00	300,00	9500,00	0,00	0,00	43100,00	0,00	13300,00	1200,00	1200,00	38200,00	1200,00	15700,00	1200,00	1200,00	21100,00	1200,00	10
Desired supply Wholesale	1500	Unit	11	0,00	900,00	300,00	10200,00	300,00	300,00	52600,00	300,00	14800,00	2700,00	2700,00	48800,00	2700,00	17800,00	1200,00	1200,00	31400,00	1200,00	11
			12	0,00	600,00	500,00	11400,00	600,00	600,00	62500,00	600,00	16300,00	400,00	400,00	59400,00	400,00	19900,00	1200,00	1200,00	43800,00	1200,00	12
Initial supply Regional Warehouse	1800	Unit	13	300,00	400,00	300,00	12600,00	1200,00	1200,00	73300,00	1200,00	17800,00	800,00	800,00	73800,00	800,00	22000,00	1200,00	1200,00	58300,00	1200,00	13
Delivery time to Regional Warehouse	4	Week	14	600,00	700,00	300,00	13500,00	2700,00	2700,00	84700,00	2700,00	19300,00	1200,00	1200,00	89300,00	1200,00	24100,00	1200,00	1200,00	74900,00	1200,00	14
Time to fill oder for to Regional Warehouse	3	Week	15	1200,00	1600,00	300,00	13500,00	400,00	400,00	95500,00	400,00	20800,00	1200,00	1200,00	105900,00	1200,00	26200,00	1200,00	1200,00	93600,00	1200,00	15
Desired supply Regional Warehouse	2100	Unit	16	2700,00	4000,00	500,00	11300,00	800,00	800,00	108600,00	800,00	22300,00	1200,00	1200,00	124000,00	1200,00	28300,00	1200,00	1200,00	114400,00	1200,00	16
			17	400,00	3900,00	300,00	9000,00	1200,00	1200,00	119100,00	1200,00	23800,00	1200,00	1200,00	143600,00	1200,00	30400,00	1200,00	1200,00	137300,00	1200,00	17
Initial supply Factory	2400	Unit	18	800,00	4400,00	300,00	6200,00	1200,00	1200,00	126900,00	1200,00	25300,00	1200,00	1200,00	164700,00	1200,00	32500,00	1200,00	1200,00	162300,00	1200,00	18
Average production Factory	800	Unit/ Week	19	1200,00	5300,00	300,00	2500,00	1200,00	1200,00	131900,00	1200,00	26800,00	1200,00	1200,00	187300,00	1200,00	34600,00	1200,00	1200,00	189400,00	1200,00	19
Max supply Factory	10000	Unit	20	1200,00	6200,00	500,00	0,00	1200,00	1200,00	133200,00	1200,00	28300,00	1200,00	1200,00	211400,00	1200,00	36700,00	1200,00	1200,00	218600,00	1200,00	20
			21	1200,00	6900,00	300,00	0,00	1200,00	1200,00	132000,00	1200,00	29800,00	1200,00	1200,00	237000,00	1200,00	38800,00	1200,00	1200,00	249900,00	1200,00	21
			22	1200,00	7800,00	300,00	0,00	1200,00	1200,00	130800,00	1200,00	31300,00	1200,00	1200,00	264100,00	1200,00	40900,00	1200,00	1200,00	283300,00	1200,00	22
			23	1200,00	8700,00	300,00	0,00	1200,00	1200,00	129600,00	1200,00	32800,00	1200,00	1200,00	292700,00	1200,00	43000,00	1200,00	1200,00	318800,00	1200,00	23
			24	1200,00	9600,00	500,00	0,00	1200,00	1200,00	128400,00	1200,00	34300,00	1200,00	1200,00	322800,00	1200,00	45100,00	1200,00	1200,00	356400,00	1200,00	24
			25	1200,00	10300,00	300,00	0,00	1200,00	1200,00	127200,00	1200,00	35800,00	1200,00	1200,00	354400,00	1200,00	47200,00	1200,00	1200,00	396100,00	1200,00	25
			26	1200,00	11200,00	300,00	0,00	1200,00	1200,00	126000,00	1200,00	37300,00	1200,00	1200,00	387500,00	1200,00	49300,00	1200,00	1200,00	437900,00	1200,00	26
			27	1200,00	12100,00	300,00	0,00	1200,00	1200,00	124800,00	1200,00	38800,00	1200,00	1200,00	422100,00	1200,00	51400,00	1200,00	1200,00	481800,00	1200,00	27
			28	1200,00	13000,00	500,00	0,00	1200,00	1200,00	123600,00	1200,00	40300,00	1200,00	1200,00	458200,00	1200,00	53500,00	1200,00	1200,00	527800,00	1200,00	28
			29	1200,00	13700,00	300,00	0,00	1200,00	1200,00	122400,00	1200,00	41800,00	1200,00	1200,00	495800,00	1200,00	55600,00	1200,00	1200,00	575900,00	1200,00	29
			30	1200,00	14600,00	300,00	0,00	1200,00	1200,00	121200,00	1200,00	43300,00	1200,00	1200,00	534900,00	1200,00	57700,00	1200,00	1200,00	626100,00	1200,00	30

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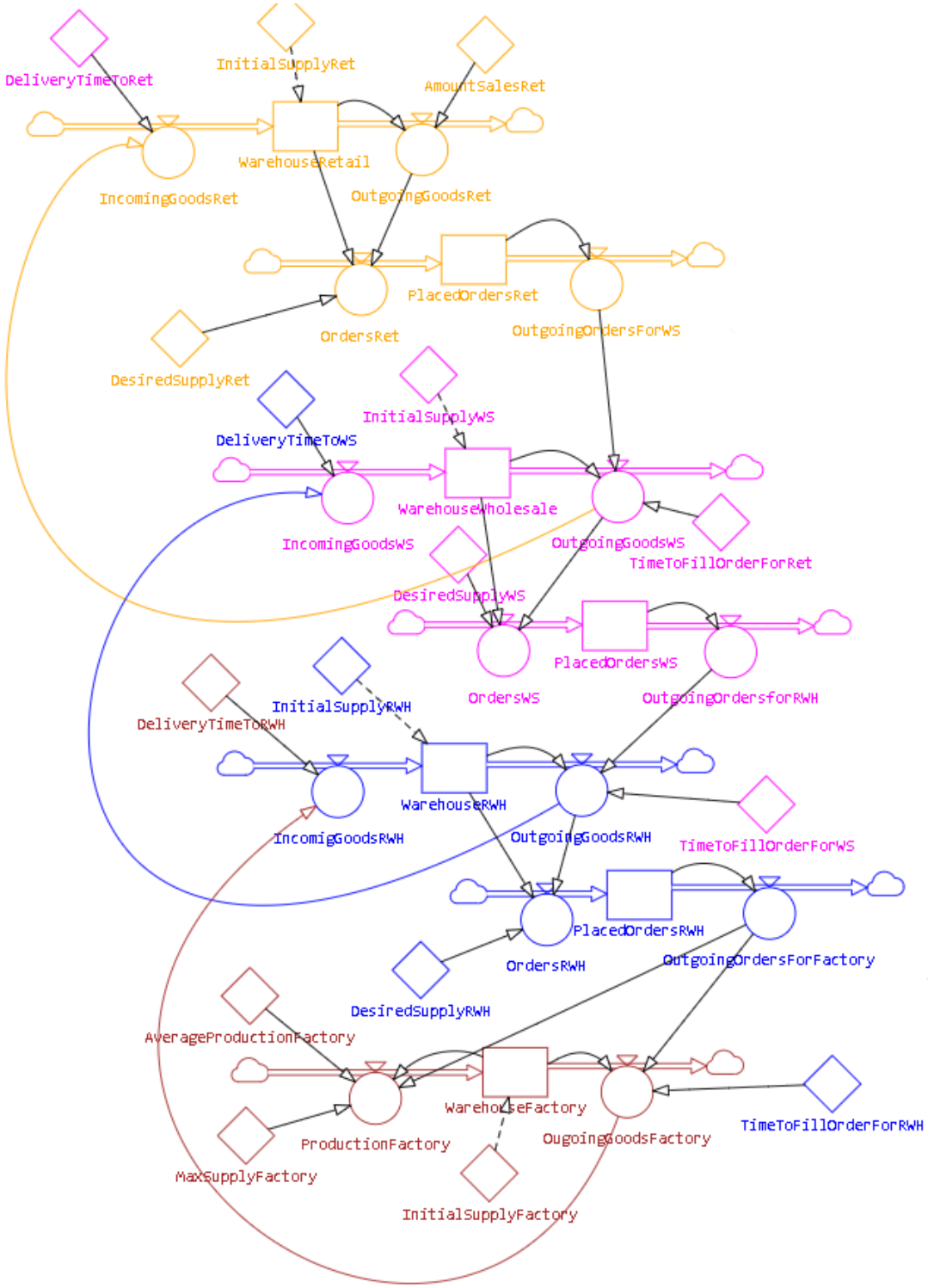
3.2. StochSD Model

To build a model within the system dynamics paradigm, the StochSD software (based on Insight Maker) was chosen. The Beer Game model with all four groups is shown in Figure 4.

Figure 4: Beer Game model in StochSD



The computation results for the first 30 weeks are presented in Figure 5.

Figure 5. Beer Game calculation in StochSD

Model SSD_1 w/o limits				Incoming goods Retail, Unit/ Week	Warehouse Retail, Unit	Outgoing goods Retail, Unit/ Week	Orders Retail, Unit/ Week	Placed orders Retail, Unit	Outgoing orders Retail, Unit/ Week	Incoming goods Wholesale, Unit/ Week	Warehouse Wholesale, Unit	Outgoing goods Wholesale, Unit/ Week	Orders Wholesale, Unit/ Week	Placed orders Wholesale, Unit	Outgoing orders Wholesale, Unit/ Week	Incoming goods RWH, Unit/ Week	Warehouse RWH, Unit	Outgoing goods RWH, Unit/ Week	Orders RWH, Unit/ Week	Placed orders RWH, Unit	Outgoing orders RWH, Unit/ Week	Production Factory, Unit/ Week	Warehouse Factory, Unit	Outgoing goods Factory, Unit/ Week	
TIME STEP	1	Week	0	0,00	900,00	0,00	0,00	0,00	0,00	0,00	1200,00	0,00	0,00	0,00	0,00	0,00	1800,00	0,00	0,00	0,00	0,00	0,00	2400,00	0,00	0
			1	0,00	900,00	300,00	700,00	0,00	0,00	0,00	1200,00	0,00	300,00	0,00	0,00	0,00	1800,00	0,00	300,00	0,00	0,00	400,00	2400,00	0,00	1
Initial supply Retail	900	Unit	2	0,00	600,00	300,00	1000,00	700,00	700,00	0,00	1200,00	0,00	300,00	300,00	300,00	0,00	1800,00	0,00	300,00	300,00	300,00	400,00	2800,00	0,00	2
Delivery time to Retail	2	Week	3	0,00	300,00	300,00	1300,00	1000,00	1000,00	0,00	1200,00	700,00	1000,00	300,00	300,00	0,00	1800,00	0,00	300,00	300,00	300,00	400,00	3200,00	0,00	3
Average amount of sales Retail	300	Unit/ Week	4	0,00	0,00	0,00	1300,00	1300,00	1300,00	0,00	500,00	500,00	1500,00	1000,00	1000,00	0,00	1800,00	300,00	600,00	300,00	300,00	400,00	3600,00	0,00	4
Time to fill oder for Retail	1	Week	5	700,00	0,00	0,00	1300,00	1300,00	1300,00	0,00	0,00	0,00	1500,00	1500,00	1500,00	0,00	1500,00	300,00	900,00	600,00	600,00	400,00	4000,00	300,00	5
Desired supply Retail	1300	Unit	6	500,00	700,00	300,00	900,00	1300,00	1300,00	0,00	0,00	0,00	1500,00	1500,00	1500,00	0,00	1200,00	1000,00	1900,00	900,00	900,00	400,00	4100,00	300,00	6
			7	0,00	900,00	300,00	700,00	900,00	900,00	300,00	0,00	0,00	1500,00	1500,00	1500,00	0,00	200,00	200,00	2100,00	1900,00	1900,00	400,00	4200,00	300,00	7
Initial supply Wholesale	1200	Unit	8	0,00	600,00	500,00	1200,00	700,00	700,00	300,00	300,00	300,00	1500,00	1500,00	1500,00	0,00	0,00	0,00	2100,00	2100,00	2100,00	400,00	4300,00	600,00	8
Delivery time to Wholesale	3	Week	9	0,00	100,00	100,00	1300,00	1200,00	1200,00	1000,00	300,00	300,00	1500,00	1500,00	1500,00	300,00	0,00	0,00	2100,00	2100,00	2100,00	800,00	4100,00	900,00	9
Time to fill oder for Wholesale	2	Week	10	300,00	0,00	0,00	1300,00	1300,00	1300,00	200,00	1000,00	1000,00	1500,00	1500,00	1500,00	300,00	300,00	300,00	2100,00	2100,00	2100,00	800,00	4000,00	1900,00	10
Desired supply Wholesale	1500	Unit	11	300,00	300,00	300,00	1300,00	1300,00	1300,00	0,00	200,00	200,00	1500,00	1500,00	1500,00	300,00	300,00	300,00	2100,00	2100,00	2100,00	800,00	2900,00	2100,00	11
			12	1000,00	300,00	300,00	1300,00	1300,00	1300,00	0,00	0,00	0,00	1500,00	1500,00	1500,00	600,00	300,00	300,00	2100,00	2100,00	2100,00	800,00	1600,00	1600,00	12
Initial supply Regional Warehouse	1800	Unit	13	200,00	1000,00	300,00	600,00	1300,00	1300,00	300,00	0,00	0,00	1500,00	1500,00	1500,00	900,00	600,00	600,00	2100,00	2100,00	2100,00	1200,00	800,00	800,00	13
Delivery time to Regional Warehouse	4	Week	14	0,00	900,00	300,00	700,00	600,00	600,00	300,00	300,00	300,00	1500,00	1500,00	1500,00	1900,00	900,00	900,00	2100,00	2100,00	2100,00	800,00	1200,00	1200,00	14
Time to fill oder for to Regional Warehouse	3	Week	15	0,00	600,00	300,00	1000,00	700,00	700,00	300,00	300,00	300,00	1500,00	1500,00	1500,00	2100,00	1900,00	1500,00	1700,00	2100,00	2100,00	1200,00	800,00	800,00	15
Desired supply Regional Warehouse	2100	Unit	16	300,00	300,00	300,00	1300,00	1000,00	1000,00	600,00	300,00	300,00	1500,00	1500,00	1500,00	1600,00	2500,00	1500,00	1100,00	1700,00	1700,00	800,00	1200,00	1200,00	16
			17	300,00	300,00	300,00	1300,00	1300,00	1300,00	900,00	600,00	600,00	1500,00	1500,00	1500,00	800,00	2600,00	1500,00	1000,00	1100,00	1100,00	800,00	800,00	800,00	17
Initial supply Factory	2400	Unit	18	300,00	300,00	300,00	1300,00	1300,00	1300,00	1500,00	900,00	900,00	1500,00	1500,00	1500,00	1200,00	1900,00	1500,00	1700,00	1000,00	1000,00	800,00	800,00	800,00	18
Average production Factory	800	Unit/ Week	19	600,00	300,00	300,00	1300,00	1300,00	1300,00	1500,00	1500,00	1300,00	1300,00	1500,00	1500,00	800,00	1600,00	1500,00	2000,00	1700,00	1700,00	1200,00	800,00	800,00	19
Max supply Factory	10000	Unit	20	900,00	600,00	500,00	1200,00	1300,00	1300,00	1500,00	1700,00	1300,00	1100,00	1300,00	1300,00	1200,00	900,00	900,00	2100,00	2000,00	2000,00	800,00	1200,00	1100,00	20
			21	1300,00	1000,00	300,00	600,00	1200,00	1200,00	1500,00	1900,00	1300,00	900,00	1100,00	1100,00	800,00	1200,00	1200,00	2100,00	2100,00	2100,00	1200,00	900,00	900,00	21
			22	1300,00	2000,00	300,00	0,00	600,00	600,00	1500,00	2100,00	1200,00	600,00	900,00	900,00	800,00	800,00	800,00	2100,00	2100,00	2100,00	800,00	1200,00	1200,00	22
			23	1300,00	3000,00	300,00	0,00	0,00	0,00	900,00	2400,00	600,00	0,00	600,00	600,00	800,00	800,00	800,00	2100,00	2100,00	2100,00	1200,00	800,00	800,00	23
			24	1200,00	4000,00	500,00	0,00	0,00	0,00	1200,00	2700,00	0,00	0,00	0,00	0,00	1100,00	800,00	800,00	2100,00	2100,00	2100,00	800,00	1200,00	1200,00	24
			25	600,00	4700,00	300,00	0,00	0,00	0,00	800,00	3900,00	0,00	0,00	0,00	0,00	900,00	1100,00	600,00	1600,00	2100,00	2100,00	1200,00	800,00	800,00	25
			26	0,00	5000,00	300,00	0,00	0,00	0,00	800,00	4700,00	0,00	0,00	0,00	0,00	1200,00	1400,00	0,00	700,00	1600,00	1600,00	800,00	1200,00	1200,00	26
			27	0,00	4700,00	300,00	0,00	0,00	0,00	800,00	5500,00	0,00	0,00	0,00	0,00	800,00	2600,00	0,00	0,00	700,00	700,00	800,00	800,00	800,00	27
			28	0,00	4400,00	500,00	0,00	0,00	0,00	600,00	6300,00	0,00	0,00	0,00	0,00	1200,00	3400,00	0,00	0,00	0,00	0,00	400,00	800,00	800,00	28
			29	0,00	3900,00	300,00	0,00	0,00	0,00	0,00	6900,00	0,00	0,00	0,00	0,00	800,00	4600,00	0,00	0,00	0,00	0,00	400,00	400,00	400,00	29
			30	0,00	3600,00	300,00	0,00	0,00	0,00	0,00	6900,00	0,00	0,00	0,00	0,00	1200,00	5400,00	0,00	0,00	0,00	0,00	400,00	400,00	400,00	30

A comparison of the results of the two models, using the Retail group as an example, is shown in Figure 6.

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3.3. Comparison of Model Results

Figure 6. Comparison of results (Retail group)

W/o limits	Incoming goods Retail, Unit/ Week		Warehouse Retail, Unit		Outgoing goods Retail, Unit/ Week		Orders Retail, Unit/ Week		Incoming goods Wholesale, Unit/ Week		Warehouse Wholesale, Unit		Placed orders Retail, Unit		Outgoing goods Wholesale, Unit/ Week		Orders Wholesale, Unit/ Week	
	SSD / Excel		SSD / Excel		SSD / Excel		SSD / Excel		SSD / Excel		SSD / Excel		SSD / Excel		SSD / Excel		SSD / Excel	
0	0,00	0,00	900,00	900,00	0,00	0,00	0,00	0,00	0,00	0,00	1200,00	1200,00	0,00	0,00	0,00	0,00	0,00	0,00
1	0,00	0,00	900,00	900,00	300,00	300,00	700,00	700,00	0,00	0,00	1200,00	1200,00	0,00	0,00	0,00	0,00	300,00	300,00
2	0,00	0,00	600,00	600,00	300,00	300,00	1000,00	1700,00	0,00	0,00	1200,00	1200,00	700,00	700,00	0,00	700,00	300,00	1300,00
3	0,00	0,00	300,00	300,00	300,00	300,00	1300,00	3000,00	0,00	0,00	1200,00	500,00	1000,00	1700,00	700,00	500,00	1000,00	2800,00
4	0,00	700,00	0,00	700,00	0,00	500,00	1300,00	4100,00	0,00	0,00	500,00	0,00	1300,00	4200,00	500,00	0,00	1500,00	4300,00
5	700,00	500,00	0,00	700,00	0,00	300,00	1300,00	5000,00	0,00	0,00	0,00	0,00	1300,00	8300,00	0,00	0,00	1500,00	5800,00
6	500,00	0,00	700,00	400,00	300,00	300,00	900,00	6200,00	0,00	300,00	0,00	300,00	1300,00	13300,00	0,00	300,00	1500,00	7300,00
7	0,00	0,00	900,00	100,00	300,00	100,00	700,00	7500,00	300,00	1300,00	0,00	1300,00	900,00	19200,00	0,00	1300,00	1500,00	8800,00
8	0,00	300,00	600,00	300,00	500,00	300,00	1200,00	8800,00	300,00	200,00	300,00	200,00	700,00	25400,00	300,00	200,00	1500,00	10300,00
9	0,00	1300,00	100,00	1300,00	100,00	300,00	1300,00	9100,00	1000,00	0,00	300,00	0,00	1200,00	34000,00	300,00	0,00	1500,00	11800,00
10	300,00	200,00	0,00	1200,00	0,00	300,00	1300,00	9500,00	200,00	0,00	1000,00	0,00	1300,00	43100,00	1000,00	0,00	1500,00	13300,00
11	300,00	0,00	300,00	900,00	300,00	300,00	1300,00	10200,00	0,00	300,00	200,00	300,00	1300,00	52600,00	200,00	300,00	1500,00	14800,00
12	1000,00	0,00	300,00	600,00	300,00	500,00	1300,00	11400,00	0,00	600,00	0,00	600,00	1300,00	62500,00	0,00	600,00	1500,00	16300,00
13	200,00	300,00	1000,00	400,00	300,00	300,00	600,00	12600,00	300,00	1200,00	0,00	1200,00	1300,00	73300,00	0,00	1200,00	1500,00	17800,00
14	0,00	600,00	900,00	700,00	300,00	300,00	700,00	13500,00	300,00	2700,00	300,00	2700,00	600,00	84700,00	300,00	2700,00	1500,00	19300,00
15	0,00	1200,00	600,00	1600,00	300,00	300,00	1000,00	13500,00	300,00	400,00	300,00	400,00	700,00	95500,00	300,00	400,00	1500,00	20800,00
16	300,00	2700,00	300,00	4000,00	300,00	500,00	1300,00	11300,00	600,00	800,00	300,00	800,00	1000,00	108600,00	300,00	800,00	1500,00	22300,00
17	300,00	400,00	300,00	3900,00	300,00	300,00	1300,00	9000,00	900,00	1200,00	600,00	1200,00	1300,00	119100,00	600,00	1200,00	1500,00	23800,00
18	300,00	800,00	300,00	4400,00	300,00	300,00	1300,00	6200,00	1500,00	1200,00	900,00	1200,00	1300,00	126900,00	900,00	1200,00	1500,00	25300,00
19	600,00	1200,00	300,00	5300,00	300,00	300,00	1300,00	2500,00	1500,00	1200,00	1500,00	1200,00	1300,00	131900,00	1300,00	1200,00	1300,00	26800,00
20	900,00	1200,00	600,00	6200,00	500,00	500,00	1200,00	0,00	1500,00	1200,00	1700,00	1200,00	1300,00	133200,00	1300,00	1200,00	1100,00	28300,00
21	1300,00	1200,00	1000,00	6900,00	300,00	300,00	600,00	0,00	1500,00	1200,00	1900,00	1200,00	1200,00	132000,00	1300,00	1200,00	900,00	29800,00
22	1300,00	1200,00	2000,00	7800,00	300,00	300,00	0,00	0,00	1500,00	1200,00	2100,00	1200,00	600,00	130800,00	1200,00	1200,00	600,00	31300,00
23	1300,00	1200,00	3000,00	8700,00	300,00	300,00	0,00	0,00	900,00	1200,00	2400,00	1200,00	0,00	129600,00	600,00	1200,00	0,00	32800,00
24	1200,00	1200,00	4000,00	9600,00	500,00	500,00	0,00	0,00	1200,00	1200,00	2700,00	1200,00	0,00	128400,00	0,00	1200,00	0,00	34300,00
25	600,00	1200,00	4700,00	10300,00	300,00	300,00	0,00	0,00	800,00	1200,00	3900,00	1200,00	0,00	127200,00	0,00	1200,00	0,00	35800,00
26	0,00	1200,00	5000,00	11200,00	300,00	300,00	0,00	0,00	800,00	1200,00	4700,00	1200,00	0,00	126000,00	0,00	1200,00	0,00	37300,00
27	0,00	1200,00	4700,00	12100,00	300,00	300,00	0,00	0,00	800,00	1200,00	5500,00	1200,00	0,00	124800,00	0,00	1200,00	0,00	38800,00
28	0,00	1200,00	4400,00	13000,00	500,00	500,00	0,00	0,00	600,00	1200,00	6300,00	1200,00	0,00	123600,00	0,00	1200,00	0,00	40300,00
29	0,00	1200,00	3900,00	13700,00	300,00	300,00	0,00	0,00	0,00	1200,00	6900,00	1200,00	0,00	122400,00	0,00	1200,00	0,00	41800,00
30	0,00	1200,00	3600,00	14600,00	300,00	300,00	0,00	0,00	0,00	1200,00	6900,00	1200,00	0,00	121200,00	0,00	1200,00	0,00	43300,00

The method for calculating warehouse supply and sales in the Excel and StochSD models is the same. Differences begin in the accumulation of orders.

The StochSD model does not accumulate orders in the queue. The order at each step is recomputed based on the desired supply and the current warehouse supply of the Retail group. Furthermore, orders are transferred to sales (outgoing goods) only in the next step, passing through the order queue level. As a result, goods arrive at the Retail group's warehouse one week later than in the Excel model. While such a shift is logical for a discrete model, it appears unnatural from the perspective of a real process.

Let's examine the dynamics of three key indicators — orders (Figure 7), incoming goods (Figure 8), and warehouse supply (Figure 9) — for all four groups over the first 50 weeks (one year).

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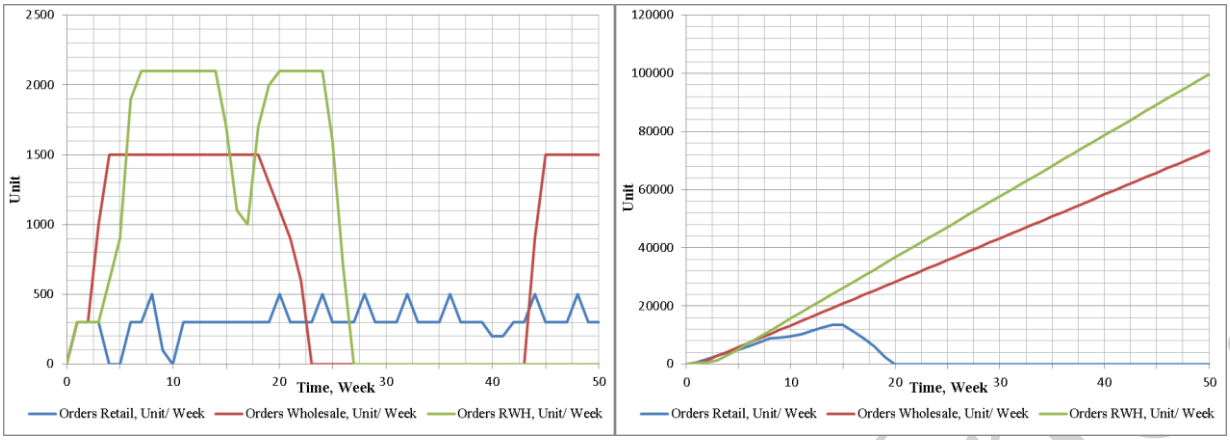
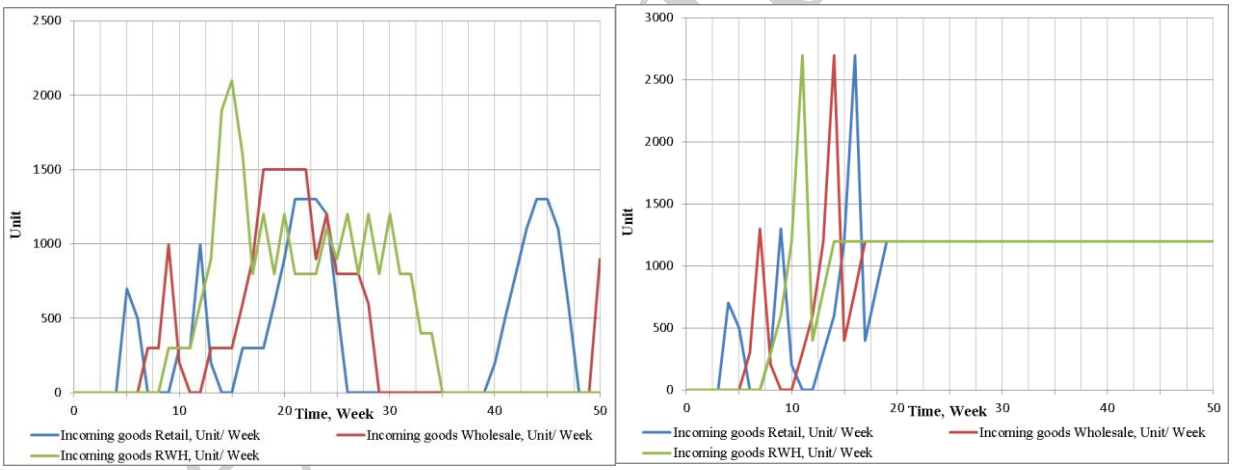
Figure 7. Orders : left – StochSD model, right – Excel model



As already noted, the StochSD model does not accumulate orders from previous steps. In contrast, the Excel computational model accounts for the fact that an ordered good does not imply immediate delivery. If the good eventually arrives at the warehouse, the need for new orders decreases automatically.

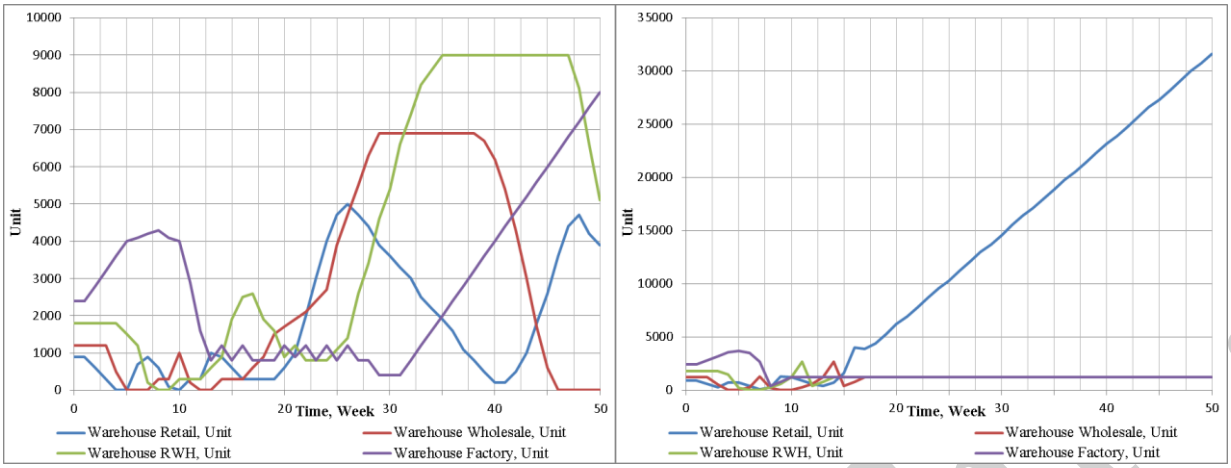
Figure 8. Incoming goods: left – StochSD model, right – Excel model



The StochSD model delivers exactly what was ordered, without considering the queue or correcting for partial shipments. If at step N the order exceeded the quantity in warehouse supply, only the available supply was shipped.

In the Excel model with order accumulation, by week 20, when the warehouses of the Wholesaler group, Regional Warehouse group, and Factory group become empty, they are replenished with the quantity that the factory can produce in one week — i.e., 1200 units.

Figure 9. Warehoses: left – StochSD model, right – Excel model



In the StochSD model, the warehouse is replenished by whatever amount "results" from the calculation, with no guarantee that all orders are fulfilled. In the Excel model, the warehouses of the Wholesale group, Regional Warehouse group, and Factory group are replenished by only 1200 units (the factory's maximum weekly output). Meanwhile, the Retail group's warehouse receives all ordered goods, leading to supply accumulation precisely in that group.

3.4. Long-Term Evaluation

Since the model describes a real supply process, it makes sense to examine the dynamics of orders (Figure 10), deliveries (Figure 11), and warehouse supply (Figure 12) over long time horizons. If a single order takes 25 weeks (half a year) to be delivered, such a supplier would not survive in the market.

For the long-term evaluation, the numerical Excel model of the Beer Game is used, as it is closer to reality and does not contain the artificial time shifts present in StochSD. The time horizon considered is 200 weeks (approximately 4 years).

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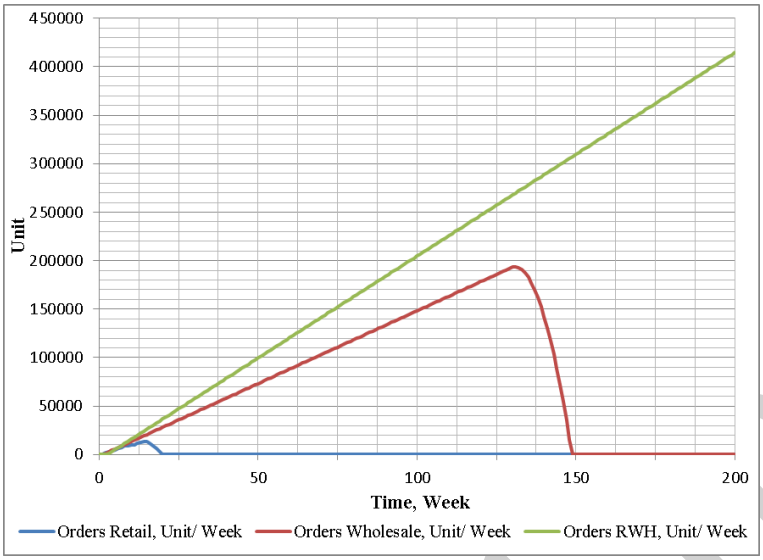
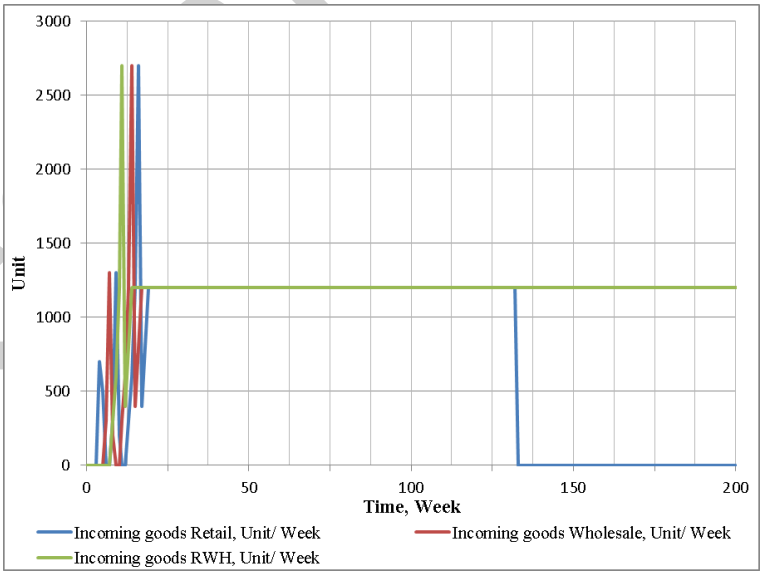
Figure 10. Order dynamics



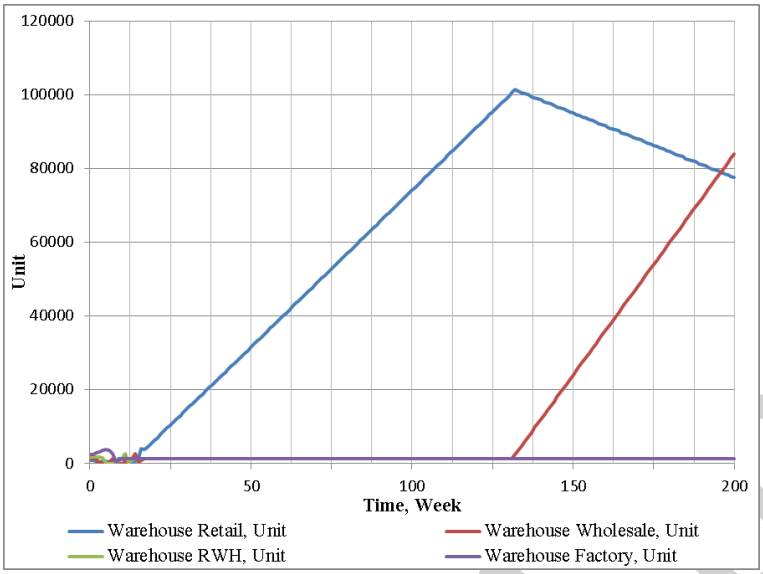
Orders from the Retail group stop at week 20. It is assumed that thereafter the warehouse will be replenished by orders that were sent to the Wholesale group during the first 20 weeks. The Wholesale group orders stop at week 150, while the Regional Warehouse group orders continue to grow.

Figure 11. Incoming goods dynamics



Goods continue to arrive at the Retail group's warehouse until week 132; after that, Retail group sales are supplied from accumulated supply (infinite warehouse). Deliveries to the Wholesale group, Regional Warehouse group, and Factory group warehouses continue at a rate equal to the Factory group's weekly production capacity.

Figure 12. Warehouse supply dynamics



Replenishment of the Retail group's warehouse from previously placed orders continues until week 132, after which the supply begins to decline due to sales. Complete depletion of the Retail group's warehouse is expected around week 400.

In reality, neither infinite warehouses nor deliveries of 200–300 units exist. However, the assumption of an infinite warehouse is one of the classical simplifications of the Beer Game, along with minimum order size for each group. Such an approach brings the model closer to reality, where no one would send half-empty trucks when 1500–2000 units can be shipped at once.

4. Beer Game Model with Minimum Order Size

The operating principle of the model remains the same, except for one condition: the Wholesale group, Regional Warehouse group, and Factory group do not ship goods from their warehouses until the accumulated order exceeds a specified minimum:

- Minimum order size of Wholesale – 1000 units;
- Minimum order size of Regional Warehouse – 1500 units;
- Minimum order size of Factory – 2000 units;

4.1. Excel Model with Minimum Order Size

Figure 13 presents the computational Beer Game model with minimum order size.

Let's trace the movement of Retail group orders:

Step 1 – the order is 700 units, which is less than the Wholesale group's minimum order size (1000). Therefore, the Retail group's order from step 1 is summed with the Retail

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group's order from step 2, giving a total of 1700 units. This total exceeds the Wholesale group's minimum order size.

The Wholesale group takes 1000 units into work and, after the Time to fill order for Retail (1 week), ships this batch to the Retail group. After the delivery time to Retail (2 weeks), the shipped batch arrives at the Retail group's warehouse.

Step 4 – the order is 4300 units, which is four times the Wholesale group's minimum order size. The Wholesale group takes 1000 units from this order and places them in the queue. The queue will continue to grow until the Wholesale group's warehouse receives a shipment from the Regional Warehouse group (Regional Warehouse group's minimum order size is 1500 units).

At step 5 the Wholesale group sends an order of 4500 units to the Regional Warehouse group. However, at step 5, the Regional Warehouse group ships only the first Wholesale group order that has reached the minimum order size (i.e., the sum of orders from the first three steps = 1900 units). The remaining 4500 units are placed in a queue and will wait until the Regional Warehouse group's warehouse is replenished by orders from the Factory.

When the Factory group has produced enough goods to ship, it will send the Regional Warehouse group its minimum order of 2000 units.

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Figure 13. Beer Game model with minimum order size in Excel

Model Excel_2 with limits					Incoming goods Retail, Unit/ Week	Warehouse Retail, Unit	Outgoing goods Retail, Unit/ Week	Orders Retail, Unit/ Week	Incoming goods Wholesale, Unit/ Week	Warehouse Wholesale, Unit	Placed orders Retail, Unit	Outgoing goods Wholesale, Unit/ Week	Orders Wholesale, Unit/ Week	Incoming goods RWH, Unit/ Week	Warehouse RWH, Unit	Outgoing orders Wholesale, Unit/ Week	Outgoing goods RWH, Unit/ Week	Orders RWH, Unit/ Week	Production Factory, Unit/ Week	Warehouse Factory, Unit	Outgoing orders RWH, Unit/ Week	Outgoing goods Factory, Unit/ Week	
TIME STEP	1	Week		0	0,00	900,00	0,00	0,00	0,00	1200,00	0,00	0,00	0,00	0,00	1800,00	0,00	0,00	0,00	0,00	2400,00	0,00	0,00	0
				1	0,00	900,00	300,00	700,00	0,00	1200,00	0,00	0,00	300,00	0,00	1800,00	0,00	0,00	300,00	400,00	2400,00	0,00	0,00	1
Initial supply Retail	900	Unit		2	0,00	600,00	300,00	1700,00	0,00	1200,00	0,00	0,00	600,00	0,00	1800,00	0,00	0,00	600,00	400,00	2800,00	0,00	0,00	2
Delivery time to Retail	2	Week		3	0,00	300,00	300,00	3000,00	0,00	1200,00	1000,00	1000,00	1900,00	0,00	1800,00	0,00	0,00	900,00	400,00	3200,00	0,00	0,00	3
Average amount of sales Retail	300	Unit/ Week		4	0,00	0,00	0,00	4300,00	0,00	200,00	1000,00	0,00	3200,00	0,00	1800,00	0,00	0,00	1200,00	400,00	3600,00	0,00	0,00	4
Time to fill oder for Retail	1	Week		5	1000,00	1000,00	300,00	4900,00	0,00	200,00	2000,00	0,00	4500,00	0,00	1800,00	1500,00	1500,00	3000,00	400,00	4000,00	0,00	0,00	5
Desired supply Retail	1300	Unit		6	0,00	700,00	300,00	5800,00	0,00	200,00	3000,00	0,00	5800,00	0,00	300,00	1500,00	0,00	4800,00	400,00	4400,00	0,00	0,00	6
				7	0,00	400,00	300,00	7000,00	0,00	200,00	4000,00	0,00	7100,00	0,00	300,00	3000,00	0,00	6600,00	400,00	4800,00	0,00	0,00	7
Initial supply Wholesale	1200	Unit		8	0,00	100,00	100,00	8300,00	1500,00	1700,00	5000,00	1000,00	7900,00	0,00	300,00	4500,00	0,00	8400,00	400,00	5200,00	2000,00	2000,00	8
Delivery time to Wholesale	3	Week		9	0,00	0,00	0,00	9600,00	0,00	700,00	5000,00	0,00	8700,00	0,00	300,00	6000,00	0,00	10200,00	400,00	3600,00	2000,00	2000,00	9
Time to fill oder for Wholesale	2	Week		10	1000,00	1000,00	300,00	10200,00	0,00	700,00	6000,00	0,00	9500,00	0,00	300,00	7500,00	0,00	12000,00	800,00	2000,00	2000,00	2000,00	10
Desired supply Wholesale	1500	Unit		11	0,00	700,00	300,00	11100,00	0,00	700,00	7000,00	0,00	10300,00	0,00	300,00	9000,00	0,00	13800,00	800,00	800,00	2000,00	0,00	11
Min order size of Wholesale	1000	Unit		12	0,00	400,00	300,00	12300,00	0,00	700,00	8000,00	0,00	11100,00	2000,00	2300,00	10500,00	1500,00	15100,00	1200,00	1600,00	4000,00	0,00	12
				13	0,00	100,00	100,00	13600,00	0,00	700,00	9000,00	0,00	11900,00	2000,00	2800,00	10500,00	1500,00	15900,00	1200,00	2800,00	6000,00	2000,00	13
Initial supply Regional Warehouse	1800	Unit		14	0,00	0,00	0,00	14900,00	0,00	700,00	10000,00	0,00	12700,00	2000,00	3300,00	10500,00	1500,00	16200,00	1200,00	2000,00	6000,00	2000,00	14
Delivery time to Regional Warehouse	4	Week		15	0,00	0,00	0,00	16200,00	1500,00	2200,00	11000,00	1000,00	13000,00	0,00	1800,00	10500,00	1500,00	18000,00	1200,00	1200,00	6000,00	0,00	15
Time to fill oder for to Regional Warehouse	3	Week		16	0,00	0,00	0,00	17500,00	1500,00	2700,00	11000,00	1000,00	12800,00	0,00	300,00	10500,00	0,00	19800,00	1200,00	2400,00	8000,00	2000,00	16
Desired supply Regional Warehouse	2100	Unit		17	1000,00	1000,00	300,00	18100,00	1500,00	3200,00	11000,00	1000,00	12100,00	2000,00	2300,00	12000,00	1500,00	21100,00	1200,00	1600,00	8000,00	0,00	17
Min order size of Regional Warehouse	1500	Unit		18	1000,00	1700,00	300,00	18000,00	1500,00	3700,00	11000,00	1000,00	10900,00	2000,00	2800,00	12000,00	1500,00	21900,00	1200,00	2800,00	10000,00	2000,00	18
				19	1000,00	2400,00	300,00	17200,00	0,00	2700,00	11000,00	1000,00	10700,00	0,00	1300,00	12000,00	0,00	22700,00	1200,00	2000,00	10000,00	2000,00	19
Initial supply Factory	2400	Unit		20	1000,00	3100,00	500,00	15900,00	1500,00	3200,00	11000,00	1000,00	10000,00	2000,00	3300,00	13500,00	1500,00	23000,00	1200,00	1200,00	10000,00	0,00	20
Average production Factory	800	Unit/ Week		21	1000,00	3600,00	300,00	13900,00	1500,00	3700,00	11000,00	1000,00	8800,00	0,00	1800,00	13500,00	1500,00	24800,00	1200,00	2400,00	12000,00	2000,00	21
Min order size of Factory	2000	Unit		22	1000,00	4300,00	300,00	11200,00	0,00	2700,00	11000,00	1000,00	8600,00	2000,00	2300,00	13500,00	1500,00	26100,00	1200,00	1600,00	12000,00	0,00	22
Max supply Factory	10000	Unit		23	1000,00	5000,00	300,00	7800,00	1500,00	3200,00	11000,00	1000,00	7900,00	2000,00	2800,00	13500,00	1500,00	26900,00	1200,00	2800,00	14000,00	2000,00	23
				24	1000,00	5700,00	500,00	3900,00	1500,00	3700,00	11000,00	1000,00	6700,00	0,00	1300,00	13500,00	0,00	27700,00	1200,00	2000,00	14000,00	2000,00	24
				25	1000,00	6200,00	300,00	0,00	1500,00	4200,00	11000,00	1000,00	5000,00	2000,00	3300,00	15000,00	1500,00	28000,00	1200,00	1200,00	14000,00	0,00	25
				26	1000,00	6900,00	300,00	0,00	1500,00	4700,00	10000,00	1000,00	2800,00	0,00	1800,00	15000,00	1500,00	29800,00	1200,00	2400,00	16000,00	2000,00	26
				27	1000,00	7600,00	300,00	0,00	0,00	3700,00	9000,00	1000,00	1600,00	2000,00	2300,00	15000,00	1500,00	31100,00	1200,00	1600,00	16000,00	0,00	27
				28	1000,00	8300,00	500,00	0,00	1500,00	4200,00	8000,00	1000,00	0,00	2000,00	2800,00	15000,00	1500,00	31900,00	1200,00	2800,00	18000,00	2000,00	28
				29	1000,00	8800,00	300,00	0,00	1500,00	4700,00	7000,00	1000,00	0,00	0,00	1300,00	15000,00	0,00	32700,00	1200,00	2000,00	18000,00	2000,00	29
				30	1000,00	9500,00	300,00	0,00	1500,00	5200,00	6000,00	1000,00	0,00	2000,00	3300,00	15000,00	1500,00	33000,00	1200,00	1200,00	18000,00	0,00	30

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4.2. StochSD Model with Minimum Order Size

The logic of the StochSD model (Figure 14) remains unchanged. The only modification is the addition of minimum order size parameters for the Wholesale group, Regional Warehouse group, and Factory group. The computation results of the StochSD model with minimum order size are shown in Figure 15.

Figure 14 Beer Game model with minimum order size in StochSD

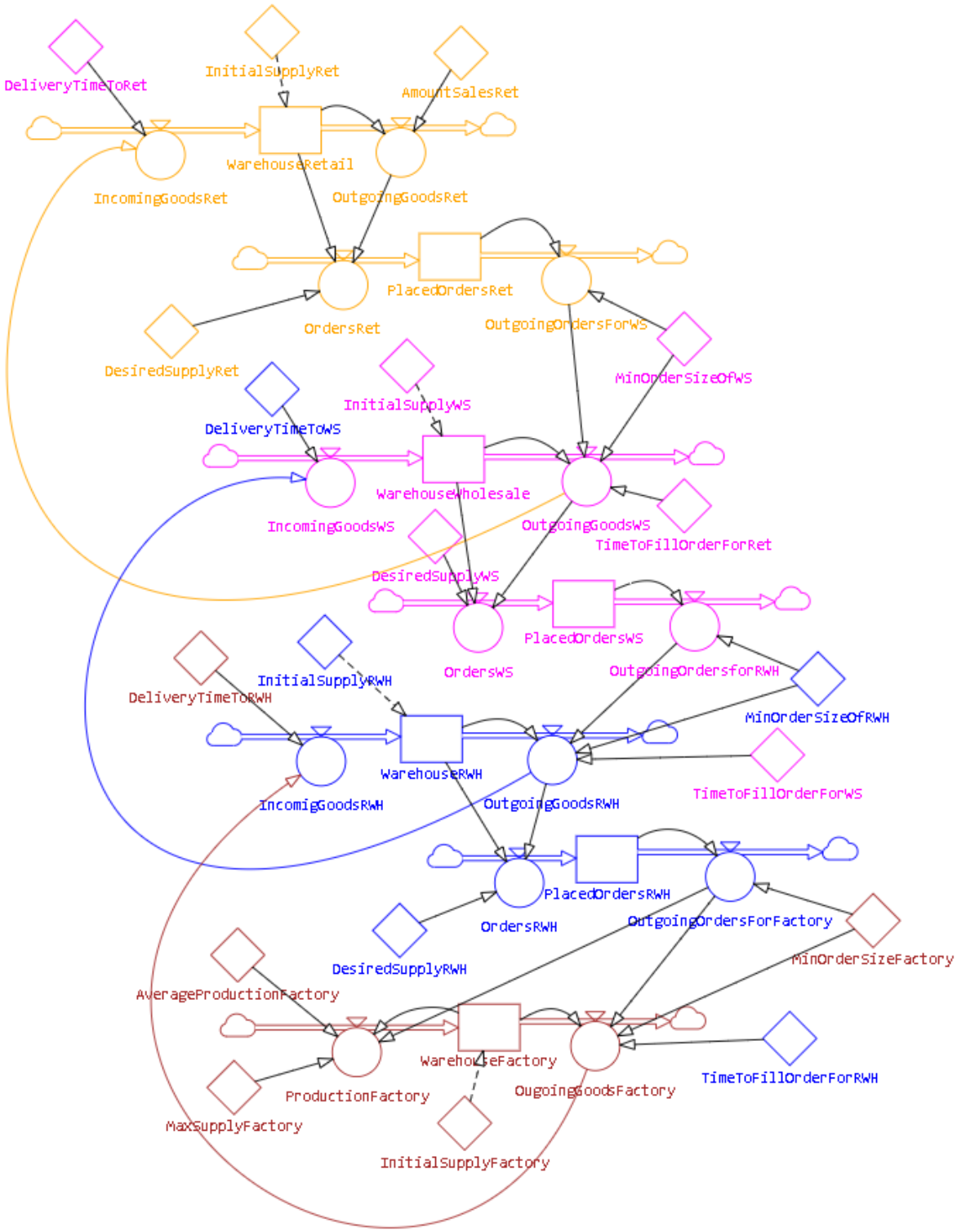

Figure 15. Calculation of the StochSD model with minimum order size

Model SSD_2 with limits				Incoming goods Retail, Unit/ Week	Warehouse Retail, Unit	Outgoing goods Retail, Unit/ Week	Orders Retail, Unit/ Week	Placed orders Retail, Unit	Outgoing orders Retail, Unit/ Week	Incoming goods Wholesale, Unit/ Week	Warehouse Wholesale, Unit	Outgoing goods Wholesale, Unit/ Week	Orders Wholesale, Unit/ Week	Placed orders Wholesale, Unit	Outgoing orders Wholesale, Unit/ Week	Incoming goods RWH, Unit/ Week	Warehouse RWH, Unit	Outgoing goods RWH, Unit/ Week	Orders RWH, Unit/ Week	Placed orders RWH, Unit	Outgoing orders RWH, Unit/ Week	Production Factory, Unit/ Week	Warehouse Factory, Unit	Outgoing goods Factory, Unit/ Week	
TIME STEP	1	Week	0	0,00	900,00	0,00	0,00	0,00	0,00	0,00	1200,00	0,00	0,00	0,00	0,00	0,00	1800,00	0,00	0,00	0,00	0,00	0,00	2400,00	0,00	0
			1	0,00	900,00	300,00	700,00	0,00	0,00	0,00	1200,00	0,00	300,00	0,00	0,00	0,00	1800,00	0,00	300,00	0,00	0,00	400,00	2400,00	0,00	1
Initial supply Retail	900	Unit	2	0,00	600,00	300,00	1000,00	700,00	0,00	0,00	1200,00	0,00	300,00	300,00	0,00	0,00	1800,00	0,00	300,00	300,00	0,00	400,00	2800,00	0,00	2
Delivery time to Retail	2	Week	3	0,00	300,00	300,00	1300,00	1700,00	1000,00	0,00	1200,00	0,00	300,00	600,00	0,00	0,00	1800,00	0,00	300,00	600,00	0,00	400,00	3200,00	0,00	3
Average amount of sales Retail	300	Unit/ Week	4	0,00	0,00	0,00	1300,00	2000,00	1000,00	0,00	1200,00	1000,00	1300,00	900,00	0,00	0,00	1800,00	0,00	300,00	900,00	0,00	400,00	3600,00	0,00	4
Time to fill oder for Retail	1	Week	5	0,00	0,00	0,00	1300,00	2300,00	1000,00	0,00	200,00	0,00	1300,00	2200,00	1500,00	0,00	1800,00	0,00	300,00	1200,00	0,00	400,00	4000,00	0,00	5
Desired supply Retail	1300	Unit	6	1000,00	0,00	0,00	1300,00	2600,00	1000,00	0,00	200,00	0,00	1300,00	2000,00	1500,00	0,00	1800,00	0,00	300,00	1500,00	0,00	400,00	4400,00	0,00	6
			7	0,00	1000,00	300,00	600,00	2900,00	1000,00	0,00	200,00	0,00	1300,00	1800,00	1500,00	0,00	1800,00	1500,00	1800,00	1800,00	0,00	400,00	4800,00	0,00	7
Initial supply Wholesale	1200	Unit	8	0,00	700,00	500,00	1100,00	2500,00	1000,00	0,00	200,00	0,00	1300,00	1600,00	1500,00	0,00	300,00	0,00	1800,00	3600,00	2000,00	400,00	5200,00	0,00	8
Delivery time to Wholesale	3	Week	9	0,00	200,00	200,00	1300,00	2600,00	1000,00	0,00	200,00	0,00	1300,00	1400,00	0,00	0,00	300,00	0,00	1800,00	3400,00	2000,00	400,00	5600,00	0,00	9
Time to fill oder for Wholesale	2	Week	10	0,00	0,00	0,00	1300,00	2900,00	1000,00	1500,00	200,00	0,00	1300,00	2700,00	1500,00	0,00	300,00	0,00	1800,00	3200,00	2000,00	400,00	6000,00	0,00	10
Desired supply Wholesale	1500	Unit	11	0,00	0,00	0,00	1300,00	3200,00	1000,00	0,00	1700,00	1000,00	800,00	2500,00	1500,00	0,00	300,00	0,00	1800,00	3000,00	2000,00	400,00	6400,00	2000,00	11
Min order size of Wholesale	1000	Unit	12	0,00	0,00	0,00	1300,00	3500,00	1000,00	0,00	700,00	0,00	800,00	1800,00	1500,00	0,00	300,00	0,00	1800,00	2800,00	2000,00	400,00	4800,00	2000,00	12
			13	1000,00	0,00	0,00	1300,00	3800,00	1000,00	0,00	700,00	0,00	800,00	1100,00	0,00	0,00	300,00	0,00	1800,00	2600,00	2000,00	800,00	3200,00	2000,00	13
Initial supply Regional Warehouse	1800	Unit	14	0,00	1000,00	300,00	600,00	4100,00	1000,00	0,00	700,00	0,00	800,00	1900,00	1500,00	0,00	300,00	0,00	1800,00	2400,00	2000,00	800,00	2000,00	2000,00	14
Delivery time to Regional Warehouse	4	Week	15	0,00	700,00	300,00	900,00	3700,00	1000,00	0,00	700,00	0,00	800,00	1200,00	0,00	2000,00	300,00	0,00	1800,00	2200,00	2000,00	1200,00	800,00	0,00	15
Time to fill oder for to Regional Warehouse	3	Week	16	0,00	400,00	300,00	1200,00	3600,00	1000,00	0,00	700,00	0,00	800,00	2000,00	1500,00	2000,00	2300,00	1500,00	1300,00	2000,00	2000,00	800,00	2000,00	2000,00	16
Desired supply Regional Warehouse	2100	Unit	17	0,00	100,00	100,00	1300,00	3800,00	1000,00	0,00	700,00	0,00	800,00	1300,00	0,00	2000,00	2800,00	0,00	0,00	1300,00	0,00	400,00	800,00	0,00	17
Min order size of Regional Warehouse	1500	Unit	18	0,00	0,00	0,00	1300,00	4100,00	1000,00	0,00	700,00	0,00	800,00	2100,00	1500,00	2000,00	4800,00	1500,00	0,00	1300,00	0,00	400,00	1200,00	0,00	18
			19	0,00	0,00	0,00	1300,00	4400,00	1000,00	1500,00	700,00	0,00	800,00	1400,00	0,00	0,00	5300,00	0,00	0,00	1300,00	0,00	400,00	1600,00	0,00	19
Initial supply Factory	2400	Unit	20	0,00	0,00	0,00	1300,00	4700,00	1000,00	0,00	2200,00	1000,00	300,00	2200,00	1500,00	2000,00	5300,00	1500,00	0,00	1300,00	0,00	400,00	2000,00	0,00	20
Average production Factory	800	Unit/ Week	21	0,00	0,00	0,00	1300,00	5000,00	1000,00	1500,00	1200,00	1000,00	1300,00	1000,00	0,00	0,00	5800,00	0,00	0,00	1300,00	0,00	400,00	2400,00	0,00	21
Min order size of Factory	2000	Unit	22	1000,00	0,00	0,00	1300,00	5300,00	1000,00	0,00	1700,00	1000,00	800,00	2300,00	1500,00	0,00	5800,00	1500,00	0,00	1300,00	0,00	400,00	2800,00	0,00	22
Max supply Factory	10000	Unit	23	1000,00	1000,00	300,00	600,00	5600,00	1000,00	1500,00	700,00	0,00	800,00	1600,00	1500,00	0,00	4300,00	0,00	0,00	1300,00	0,00	400,00	3200,00	0,00	23
			24	1000,00	1700,00	500,00	100,00	5200,00	1000,00	0,00	2200,00	1000,00	300,00	900,00	0,00	0,00	4300,00	1500,00	0,00	1300,00	0,00	400,00	3600,00	0,00	24
			25	0,00	2200,00	300,00	0,00	4300,00	1000,00	1500,00	1200,00	1000,00	1300,00	1200,00	0,00	0,00	2800,00	1500,00	800,00	1300,00	0,00	400,00	4000,00	0,00	25
			26	1000,00	1900,00	300,00	0,00	3300,00	1000,00	0,00	1700,00	1000,00	800,00	2500,00	1500,00	0,00	1300,00	0,00	800,00	2100,00	2000,00	400,00	4400,00	0,00	26
			27	1000,00	2600,00	300,00	0,00	2300,00	1000,00	1500,00	700,00	0,00	800,00	1800,00	1500,00	0,00	1300,00	0,00	800,00	900,00	0,00	400,00	4800,00	0,00	27
			28	1000,00	3300,00	500,00	0,00	1300,00	1000,00	1500,00	2200,00	1000,00	300,00	1100,00	0,00	0,00	1300,00	0,00	800,00	1700,00	0,00	400,00	5200,00	0,00	28
			29	0,00	3800,00	300,00	0,00	300,00	0,00	0,00	2700,00	1000,00	0,00	1400,00	0,00	0,00	1300,00	0,00	800,00	2500,00	2000,00	400,00	5600,00	2000,00	29
			30	1000,00	3500,00	300,00	0,00	300,00	0,00	0,00	1700,00	0,00	0,00	1400,00	0,00	0,00	1300,00	0,00	800,00	1300,00	0,00	400,00	4000,00	0,00	30

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4.3. Comparison of Model Results with Minimum Order Size

A comparative table of results for the Retail group in the Excel and StochSD models is shown in Figure 16.

Figure 16. Comparative results table for the Retail group

With limits	Incoming goods Retail, Unit/ Week SSD / Excel		Warehouse Retail, Unit SSD / Excel		Outgoing goods Retail, Unit/ Week SSD / Excel		Orders Retail, Unit/ Week SSD / Excel		Incoming goods Wholesale, Unit/ Week SSD / Excel		Warehouse Wholesale, Unit SSD / Excel		Placed orders Retail, Unit SSD / Excel		Outgoing goods Wholesale, Unit/ Week SSD / Excel		Orders Wholesale, Unit/ Week SSD / Excel	
0	0,00	0,00	900,00	900,00	0,00	0,00	0,00	0,00	0,00	0,00	1200,00	1200,00	0,00	0,00	0,00	0,00	0,00	0,00
1	0,00	0,00	900,00	900,00	300,00	300,00	700,00	700,00	0,00	0,00	1200,00	1200,00	0,00	0,00	0,00	0,00	300,00	300,00
2	0,00	0,00	600,00	600,00	300,00	300,00	1000,00	1700,00	0,00	0,00	1200,00	1200,00	0,00	0,00	0,00	0,00	300,00	600,00
3	0,00	0,00	300,00	300,00	300,00	300,00	1300,00	3000,00	0,00	0,00	1200,00	1200,00	1000,00	1000,00	0,00	1000,00	300,00	1900,00
4	0,00	0,00	0,00	0,00	0,00	0,00	1300,00	4300,00	0,00	0,00	1200,00	200,00	1000,00	1000,00	1000,00	0,00	1300,00	3200,00
5	0,00	1000,00	0,00	1000,00	0,00	300,00	1300,00	4900,00	0,00	0,00	200,00	200,00	1000,00	2000,00	0,00	0,00	1300,00	4500,00
6	1000,00	0,00	0,00	700,00	0,00	300,00	1300,00	5800,00	0,00	0,00	200,00	200,00	1000,00	3000,00	0,00	0,00	1300,00	5800,00
7	0,00	0,00	1000,00	400,00	300,00	300,00	600,00	7000,00	0,00	0,00	200,00	200,00	1000,00	4000,00	0,00	0,00	1300,00	7100,00
8	0,00	0,00	700,00	100,00	500,00	100,00	1100,00	8300,00	0,00	1500,00	200,00	1700,00	1000,00	5000,00	0,00	1000,00	1300,00	7900,00
9	0,00	0,00	200,00	0,00	200,00	0,00	1300,00	9600,00	0,00	0,00	200,00	700,00	1000,00	5000,00	0,00	0,00	1300,00	8700,00
10	0,00	1000,00	0,00	1000,00	0,00	300,00	1300,00	10200,00	1500,00	0,00	200,00	700,00	1000,00	6000,00	0,00	0,00	1300,00	9500,00
11	0,00	0,00	0,00	700,00	0,00	300,00	1300,00	11100,00	0,00	0,00	1700,00	700,00	1000,00	7000,00	1000,00	0,00	800,00	10300,00
12	0,00	0,00	0,00	400,00	0,00	300,00	1300,00	12300,00	0,00	0,00	700,00	700,00	1000,00	8000,00	0,00	0,00	800,00	11100,00
13	1000,00	0,00	0,00	100,00	0,00	100,00	1300,00	13600,00	0,00	0,00	700,00	700,00	1000,00	9000,00	0,00	0,00	800,00	11900,00
14	0,00	0,00	1000,00	0,00	300,00	0,00	600,00	14900,00	0,00	0,00	700,00	700,00	1000,00	10000,00	0,00	0,00	800,00	12700,00
15	0,00	0,00	700,00	0,00	300,00	0,00	900,00	16200,00	0,00	1500,00	700,00	2200,00	1000,00	11000,00	0,00	1000,00	800,00	13000,00
16	0,00	0,00	400,00	0,00	300,00	0,00	1200,00	17500,00	0,00	1500,00	700,00	2700,00	1000,00	11000,00	0,00	1000,00	800,00	12800,00
17	0,00	1000,00	100,00	1000,00	100,00	300,00	1300,00	18100,00	0,00	1500,00	700,00	3200,00	1000,00	11000,00	0,00	1000,00	800,00	12100,00
18	0,00	1000,00	0,00	1700,00	0,00	300,00	1300,00	18000,00	0,00	1500,00	700,00	3700,00	1000,00	11000,00	0,00	1000,00	800,00	10900,00
19	0,00	1000,00	0,00	2400,00	0,00	300,00	1300,00	17200,00	1500,00	0,00	700,00	2700,00	1000,00	11000,00	0,00	1000,00	800,00	10700,00
20	0,00	1000,00	0,00	3100,00	0,00	500,00	1300,00	15900,00	0,00	1500,00	2200,00	3200,00	1000,00	11000,00	1000,00	1000,00	300,00	10000,00
21	0,00	1000,00	0,00	3600,00	0,00	300,00	1300,00	13900,00	1500,00	1500,00	1200,00	3700,00	1000,00	11000,00	1000,00	1000,00	1300,00	8800,00
22	1000,00	1000,00	0,00	4300,00	0,00	300,00	1300,00	11200,00	0,00	0,00	1700,00	2700,00	1000,00	11000,00	1000,00	1000,00	800,00	8600,00
23	1000,00	1000,00	1000,00	5000,00	300,00	300,00	600,00	7800,00	1500,00	1500,00	700,00	3200,00	1000,00	11000,00	0,00	1000,00	800,00	7900,00
24	1000,00	1000,00	1700,00	5700,00	500,00	500,00	100,00	3900,00	0,00	1500,00	2200,00	3700,00	1000,00	11000,00	1000,00	1000,00	300,00	6700,00
25	0,00	1000,00	2200,00	6200,00	300,00	300,00	0,00	0,00	1500,00	1500,00	1200,00	4200,00	1000,00	11000,00	1000,00	1000,00	1300,00	5000,00
26	1000,00	1000,00	1900,00	6900,00	300,00	300,00	0,00	0,00	0,00	1500,00	1700,00	4700,00	1000,00	10000,00	1000,00	1000,00	800,00	2800,00
27	1000,00	1000,00	2600,00	7600,00	300,00	300,00	0,00	0,00	1500,00	0,00	700,00	3700,00	1000,00	9000,00	0,00	1000,00	800,00	1600,00
28	1000,00	1000,00	3300,00	8300,00	500,00	500,00	0,00	0,00	1500,00	1500,00	2200,00	4200,00	1000,00	8000,00	1000,00	1000,00	300,00	0,00
29	0,00	1000,00	3800,00	8800,00	300,00	300,00	0,00	0,00	0,00	1500,00	2700,00	4700,00	0,00	7000,00	1000,00	1000,00	0,00	0,00
30	1000,00	1000,00	3500,00	9500,00	300,00	300,00	0,00	0,00	0,00	1500,00	1700,00	5200,00	0,00	6000,00	0,00	1000,00	0,00	0,00
31	1000,00	1000,00	4300,00	10300,00	300,00	300,00	0,00	0,00	0,00	1500,00	1700,00	5700,00	0,00	5000,00	0,00	1000,00	0,00	0,00

As can be seen from Figure 16, the absence of order accumulation in StochSD leads to distortion in the dynamics of orders. The order formula in this model considers only the current warehouse supply and the desired supply; if little goods arrive at the warehouse, orders decrease, and the bullwhip effect (in this context, the sudden filling of the warehouse with orders from the queue) is not observed.

Let's examine the dynamics of orders (Figure 17), incoming goods (Figure 18), and warehouse supply (Figure 19) for all four groups, for each model with the minimum order size constraint, over the first 50 weeks (one year).

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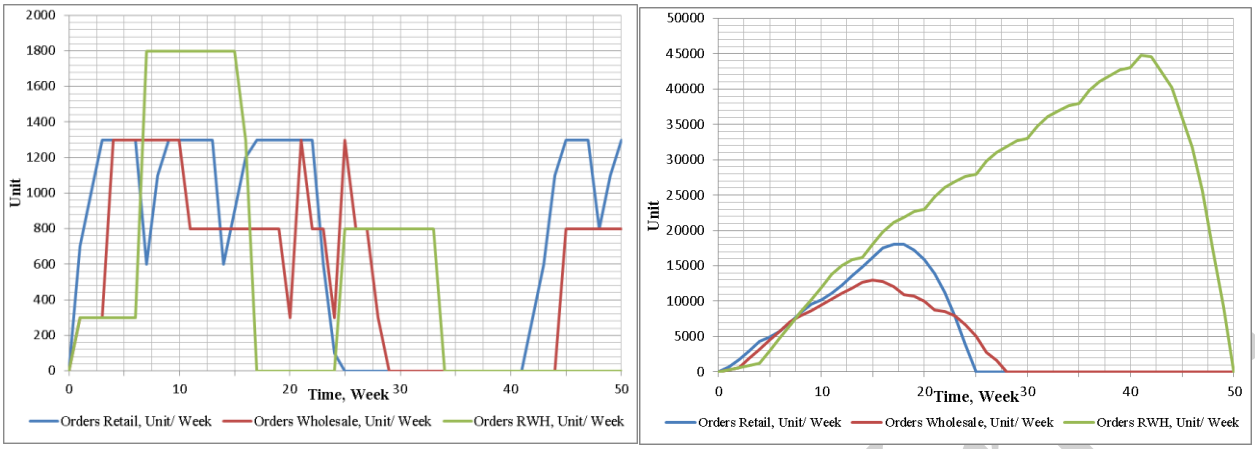
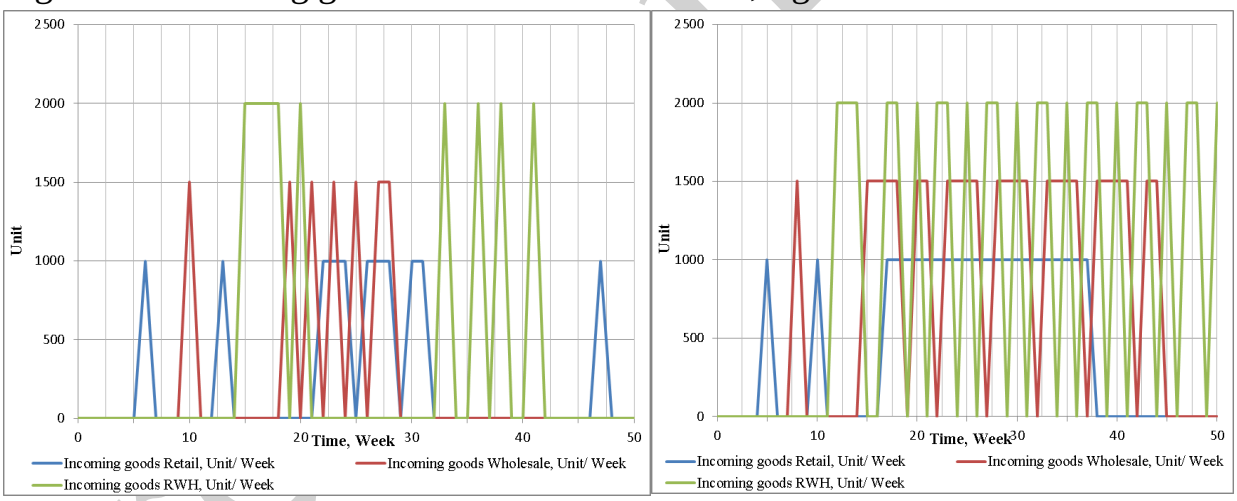
Figure 17. Orders: left – StochSD model, right – Excel model



The StochSD model still does not accumulate orders. The order at each step is determined solely by the desired supply and the actual warehouse supply. In the constrained Excel model, orders begin to accumulate significantly faster than in the unconstrained model.

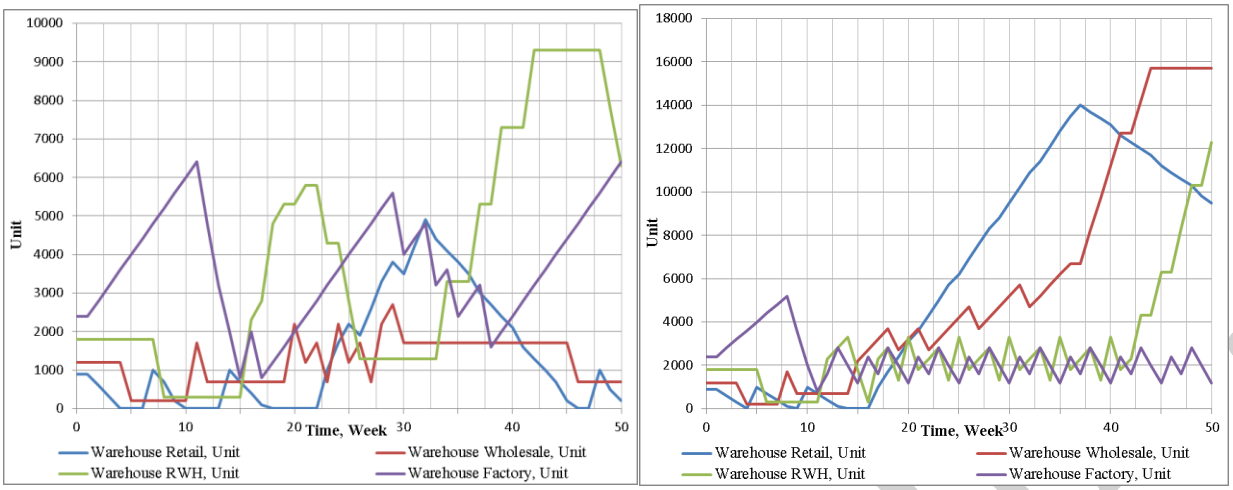
Figure 18. Incoming goods: left – StochSD model, right – Excel model



In StochSD, shipment from the Wholesalegroup's warehouse occurs only when an order calculated at the current step arrives. If the order is smaller than the minimum order size, no shipment takes place.

The Excel model behaves more realistically. After week 10, the bullwhip effect is observed: orders begin to arrive in weekly batches and continue until all accumulated queued orders have been shipped.

Figure 19. Warehouses: left – StochSD model, right – Excel model

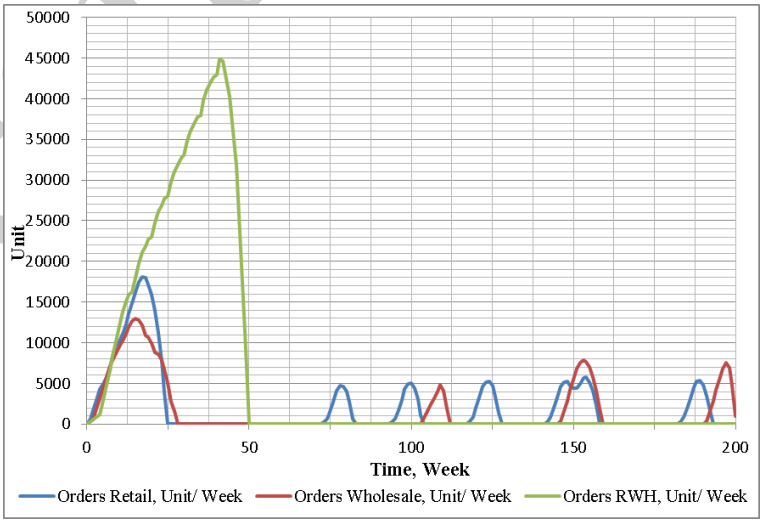


From the previous graphs, it follows that the StochSD model will replenish the warehouse only with those orders that satisfy the minimum order size requirement. In the Excel model, the warehouse is replenished by the total amount of all orders placed

4.4. Long-Term Evaluation

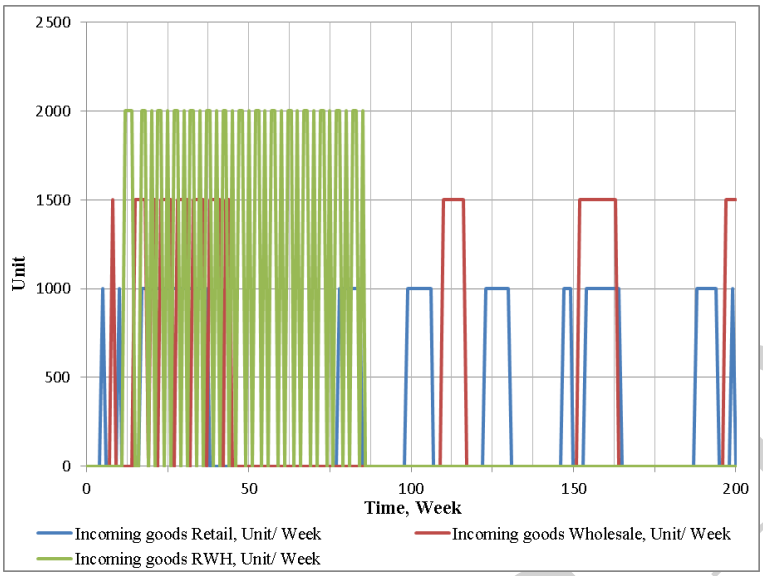
Let's examine the dynamics of orders (Figure 20), incoming goods (Figure 21), and warehouse supply (Figure 22) over a long time horizon for the Excel computational model. The time horizon considered is 200 weeks (approximately 4 years).

Figure 20. Order dynamics



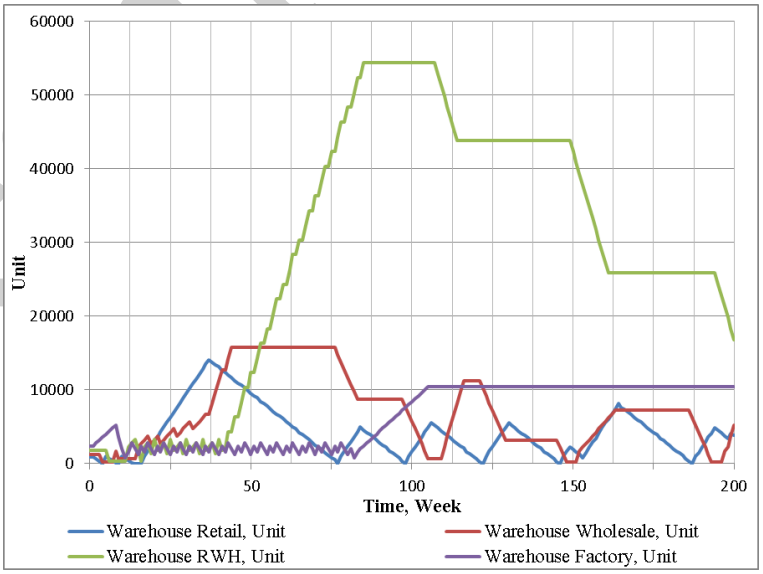
Retail group's orders stop at week 25, Wholesale group's orders at week 27, and Regional Warehouse group's orders at week 50. Further sales are supported by existing warehouse supplies. The graph also shows repeated Retail group's orders at weeks 73, 94, and 118 (roughly every 20 weeks), when the warehouses become empty. The Wholesale group repeat orders less frequently but in larger batches.

Figure 21. Incoming goods dynamics



The graph shows the bullwhip effect more clearly, with goods being delivered weekly. Most often, the Regional Warehouse group receives deliveries, followed by the Wholesale group, and then the Retail group. After week 100, repeated deliveries are visible — the graph shows how the model accounts for Time to fill order and delivery time compared to the order dynamics.

Figure 22. Warehouse dynamics



From the incoming goods dynamics graphs, it is logical that the Regional Warehouse group's warehouse is filled the most — it receives goods more frequently and in larger volumes.

At the beginning of the period, the Factory group's supply increases: production has already started, but orders from the Regional Warehouse group have not yet arrived. At

week 8, a sharp drop in the Factory group's supply occurs, accompanied by the replenishment of the Regional Warehouse group's warehouse, followed by the Wholesale group's and Retail group's warehouses. Repeated replenishment of the Factory group's warehouse begins at week 82, when the Retail group's warehouse is already empty, but deliveries continue from the supplies of the Wholesale group and Regional Warehouse group.

To prevent the Factory group from producing an infinite amount of goods, a "Max supply Fatory" parameter, set to 10,000 units, has been introduced. In addition, a reduction in the production rate is implemented when the warehouse becomes overfilled.

5. Conclusion

To compare discrete and continuous modeling approaches, two Beer Game models were built — one in Excel and one in StochSD. It was empirically found that the presence of minimum order size amplifies the bullwhip effect. To verify the model within the system dynamics paradigm, a StochSD model was constructed; its results were also recalculated in Excel for convenient comparison.

Unlike the Excel model, the StochSD model does not accumulate orders, which smooths the fluctuations in orders, incoming goods, and warehouse supply. This is due to the numerical integration features of the StochSD (and Insight Maker) environment. In contrast, the Excel model allows explicit control over the computation sequence and accounts for unfilled orders, yielding more realistic dynamics.

Therefore, for modeling supply chains with discrete batch sizes, spreadsheets are preferable to classical system dynamics environments.

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